ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

# ELECTRICAL TECHNOLOGY

## MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central Grade 10 Electrical Technology practice examination.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers.

## QUESTION 1

**1** 1.1 C ✓✓ 1.2 B ✓✓ 1.3 B ✓✓ 1.4 B ✓✓ 1.5 B ✓✓ [10]

## QUESTION 2

**2.1** Any THREE: touching live (exposed) wires ✓; damaged insulation on cables ✓; working on circuits that are switched on ✓; wet hands or wet floors; faulty appliances without earthing (3)

**2.2** Do not touch the person ✓. Switch off the power at the main switch or remove the plug, or push the person away with a dry non-conducting object (wood) ✓. Call for medical help and give first aid / CPR if trained ✓. (3)

**2.3** It detects current leaking to earth (e.g. through a person) ✓ and switches off the supply quickly to prevent electric shock and fire ✓. (2)

**2.4** Any FOUR: never work on live circuits ✓; use insulated tools ✓; keep the work area dry ✓; get the teacher to check a circuit before switching on ✓; know where the main switch and fire extinguisher are; no horseplay; report damaged equipment (4)

**[12]**

## QUESTION 3

**3.1** Side cutters: cut wires ✓; long-nose pliers: hold and bend small wires and parts in tight spaces ✓; wire strippers: remove insulation from wires ✓; soldering iron: melt solder to join components and wires ✓ (4)

**3.2** In series with the load ✓ (the circuit must be broken to insert the meter), with the dial on the current (A) range ✓. Precaution, any ONE: start on the highest range / check polarity / never connect it across the supply ✓. (3)

**3.3** Any THREE: shiny and smooth ✓; concave (volcano) shape ✓; solder flows around the whole joint ✓; no dry joints or cracks; not too much solder (3)

**3.4** $\text{Yellow } 4, \text{violet } 7$ ✓, $\text{orange} \times 10^3$ ✓ $= 47\,000\ \Omega = 47\ \text{k}\Omega, \text{gold} \pm 5\%$ ✓ (3)

**[13]**

## QUESTION 4

**4.1** The current through a conductor is directly proportional to the potential difference across it ✓, provided the temperature stays constant ✓. (2)

**4.2.1** $R_T = 10 + 20 + 30$ ✓ $= 60\ \Omega$ ✓ (2)

**4.2.2** $I = \frac{V}{R} = \frac{12}{60}$ ✓✓ $= 0{,}2\ \text{A}$ ✓ (3)

**4.2.3** $V = IR = 0{,}2 \times 20$ ✓✓ $= 4\ V$ ✓ (3)

**4.2.4** $P = VI = 12 \times 0{,}2$ ✓✓ $= 2{,}4\ W$ ✓ (3)

**4.3** $\frac{1}{R} = \frac{1}{10} + \frac{1}{10}$ ✓ $\rightarrow R = 5\ \Omega$ ✓ (2)

**4.4** The total current entering a junction equals the total current leaving the junction ✓✓. (2)

**4.5** (a) 3 V ✓ (b) 1,5 V ✓. Advantage of parallel: the cells last longer / can supply more current at the same voltage ✓✓. (4)

**4.6** Any TWO, 2 marks each: DC flows in one direction only, while AC changes direction continuously ✓✓; DC comes from cells and batteries, while AC comes from generators / the mains (230 V, 50 Hz) ✓✓; AC voltage can easily be changed with a transformer. (4)

**[25]**

## QUESTION 5

**5.1** Forward bias: the anode is connected to the positive and the cathode to the negative of the supply ✓✓. The diode conducts once the voltage is above about 0,7 V (silicon) ✓, so current flows ✓. (4)

**5.2** To limit the current ✓ so that the LED is not damaged (burnt out) ✓. (2)

**5.3** $V_R = 9 - 2 = 7\ V$ ✓; $R = \frac{7}{0{,}02}$ ✓ $= 350\ \Omega$ ✓ (3)

**5.4** It stores electric charge (energy) ✓ and releases it when needed (e.g. smoothing, timing) ✓. (2)

**5.5** As the light gets brighter, its resistance decreases ✓; in the dark, the resistance increases ✓. Use, any ONE: automatic street lights / night lights / light meters ✓. (3)

**5.6** Any TWO: single-pole single-throw (SPST) ✓; single-pole double-throw (SPDT) ✓; push-button; toggle; slide; rotary; reed switch (2)

**5.7** A fuse melts and must be replaced after it blows ✓✓, while a circuit breaker trips and can be reset. (2)

**5.8** Its resistance decreases ✓✓. (2)

**[20]**

## QUESTION 6

**6.1** $13 = 8 + 4 + 1$ ✓ $= 1101_2$ ✓ (2)

**6.2** $16 + 4 + 2$ ✓ $= 22$ ✓ (2)

**6.3** (6)

| AND | | |
|---|---|---|
| **A** | **B** | **Y** |
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

| OR | | |
|---|---|---|
| **A** | **B** | **Y** |
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

Each table: headings ✓, all input combinations ✓, correct outputs ✓

**6.4** It inverts the input ✓: an input of 1 gives an output of 0, and 0 gives 1 ✓. (2)

**6.5** $\text{AND: } Y = A \cdot B$ ✓; $\text{OR: } Y = A + B$ ✓ (2)

**6.6** NAND gate ✓. ✓✓✓ (4)

| **A** | **B** | **Y** |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**6.7** Any TWO: electronic circuits easily represent two states (on/off, high/low voltage) ✓; binary signals are less affected by noise / more reliable ✓; simple and cheap to build circuits (2)

**[20]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**